

absfigure: User Guide

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1 What the package does

absfigure lets you place figures and tables on an explicit page, at the top or bottom, and in a specific column when the document is two-column. In other words, it supports requests such as:

- put this figure at the top of page 3, left column
- put this table at the bottom of page 6, right column
- put this wide figure across both columns on page 2

Unlike absolute positioning packages that place content outside the float system, **absfigure** still uses L^AT_EX float machinery. Captions, labels, references, list-of-figures, and list-of-tables continue to work normally, and regular **figure**/**table** floats can still appear alongside absolute ones.

2 Provided environments

The package provides four environments:

Environment	Meaning
absfigure	single-column figure
absfigure*	double-column figure
abstable	single-column table
abstable*	double-column table

All four environments use the same optional key syntax:

```
\begin{absfigure}[page=3,pos=t,col=2]
...
\end{absfigure}
```

3 Quick start

```
\documentclass[twocolumn]{article}
\usepackage{graphicx}
\usepackage{absfigure}

\begin{document}
```

```

\begin{absfigure}[page=3,pos=t,col=1]
  \centering
  \includegraphics[width=\columnwidth]{fig.pdf}
  \caption{Top of page 3, left column}
  \label{fig:left}
\end{absfigure}

\begin{absfigure*}[page=4,pos=b]
  \centering
  \includegraphics[width=\linewidth]{wide.pdf}
  \caption{Bottom of page 4, full width}
  \label{fig:wide}
\end{absfigure*}

\begin{abstable}[pos=b,col=r]
  \centering
  \caption{Bottom of the current page, right column}
  \label{tab:results}
  \begin{tabular}{lc}
    \toprule
    Method & Score \\
    \midrule
    Ours & 94.2 \\
    \bottomrule
  \end{tabular}
\end{abstable}

\end{document}

```

4 Keys

Key	Values	Default	Meaning
page	integer, current, end	current	target page number
pos	t, top, b, bottom	t	top or bottom placement
col	1, l, left, 2, r, right	1	target column; ignored for starred environments

page

- Omit `page`, or write `page=current`, to target the page being built where the environment appears.
- Use `page=<number>` to target a specific page.
- Use `page=end` to append a final page containing the float.

pos

- pos=t or pos=top places the float at the top of the target page or column.
- pos=b or pos=bottom places the float at the bottom of the target page or column.

col

- col=1, col=l, or col=left selects the left column.
- col=2, col=r, or col=right selects the right column.
- For absfigure* and abstable*, the package ignores col because the float spans both columns.

5 Common patterns

Single-column figure on a specific page

```
\begin{absfigure}[page=5,pos=t,col=2]
  \centering
  \includegraphics[width=\columnwidth]{example.pdf}
  \caption{Top of page 5, right column}
\end{absfigure}
```

Wide figure across both columns

```
\begin{absfigure*}[page=2,pos=t]
  \centering
  \includegraphics[width=\linewidth]{overview.pdf}
  \caption{Spanning figure on page 2}
\end{absfigure*}
```

Inside absfigure* and abstable*, \linewidth is the full text width, as with ordinary figure* and table*.

Table on the current page

```
\begin{abstable}[pos=b,col=1]
  \centering
  \caption{Bottom of the current page}
  \begin{tabular}{lc}
    ...
  \end{tabular}
\end{abstable}
```

Appendix-style final page

```
\begin{absfigure*}[page=end,pos=t]
  \centering
  \includegraphics[width=\linewidth]{teaser.pdf}
  \caption{Placed on a final appended page}
\end{absfigure*}
```

6 How placement behaves

Ordering

If several absolute floats target the same slot, meaning the same combination of `page`, `col`, and `pos`, they are placed in source order.

Pages processed by the `balance` package are supported. Absolute floats remain attached to their requested physical columns after balancing.

Interaction with `balance` and ordinary floats

The `balance` package leaves balancing enabled after `\balance`; it remains active until `\nobalance`. If `\balance` is used before a forced page boundary to balance only the preceding material, disable it immediately after the boundary:

```
\balance
% final material on the page to balance
\newpage      % or \clearpage
\nobalance
```

Leaving `\balance` active across later pages can make ordinary `figure` and `table` floats appear out of source order. This behavior comes from `balance`, which builds and splits combined column material; it is not caused by the absolute-float environments. `absfigure` deliberately does not redefine the standard float environments or insert implicit float barriers.

The usual $\text{\LaTeX}$ rules also continue to apply to ordinary floats. In particular, a single-column `[H]` float can overtake deferred `figure*` or `table*` floats. When source order across such a semantic boundary matters, load `placeins` and insert `\FloatBarrier` explicitly at that boundary.

Reruns

If a float targets a page that has already been shipped out in the current run, $\text{\LaTeX}$ cannot go back and modify that earlier page immediately. In that case, `absfigure` saves the float definition and replays it on the next run.

When this happens, the log includes a warning telling you that placement is not yet final and that you should rerun $\text{\LaTeX}$.

This is normal in cases such as:

- a float defined on page 6 that targets page 4
- major edits that shift page breaks
- multiple absolute floats competing for the same constrained slot

What happens when a float does not fit

The package does not crop or overlap floats. If the requested page, position, and column do not have enough room, the float may remain unplaced and the log will report that fact. Typical reasons are:

- the float is taller than the available space

- too many floats are competing for the same slot
- the requested placement is structurally impossible

7 What the package loads

For normal use, add only:

```
\usepackage{absfigure}
```

`absfigure` loads its own direct dependencies. Users do not need to preload the packages it requires internally.

You still need to load content-specific packages when your float body uses them, for example:

- `graphicx` for `\includegraphics`
- `subcaption` for `subfigure`
- `booktabs`, `tabularx`, or similar table packages

8 Installation

Project-local installation

Copy `absfigure.sty` into your project directory and load it with `\usepackage{absfigure}`.

Maintainer / CTAN source layout

The canonical source files are:

- `source/latex/absfigure/absfigure.dtx`
- `source/latex/absfigure/absfigure.ins`

To regenerate the runtime package from the documented source:

```
latex source/latex/absfigure/absfigure.ins
```

9 Where to look next

The repository also includes:

- `README.md` for a short overview
- `doc/latex/absfigure/absfigure-test.tex` for a demo document
- `source/latex/absfigure/absfigure.dtx` for the package source